

William T Brown

GIS Analyst • Archaeologist • Spatial Data Specialist

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SUMMARY

GIS analyst with a Master of Science in GIS and extensive experience applying spatial technologies to archaeology, land management, and environmental compliance. Skilled in ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise, QGIS, and geodatabase management. Proficient in Python (ArcPy), R, SQL, and Esri ModelBuilder. Experienced in GPS data collection, total station mapping, cartographic production, and suitability modeling for federal and academic stakeholders.

EXPERIENCE

*University of Arizona | School of Geography and Development | **Graduate Teaching Assistant***

Jan 2025 - Apr 2025

Supported instruction for an upper-division GIS course focused on advanced geospatial techniques and enterprise GIS platforms. Graded assignments and discussion posts involving ArcGIS Pro, ArcGIS Online, and ArcGIS Enterprise workflows. Provided one-on-one technical guidance to students on spatial analysis, web mapping, and data management concepts using the Esri ecosystem.

*University of Arizona | Division for Late Medieval and Reformation Studies | **GIS Analyst***

May 2024 - Dec 2024

Developed a geospatial dataset and created an interactive historical map of Catholic and Protestant churches in the Holy Roman Empire circa 1648 using ArcGIS Pro. Responsibilities included georeferencing a historical map source, digitizing spatial features, and standardizing attribute data to support ongoing research led by Dr. Beth Plummer (Department of History/Medieval Studies). This work contributed to the Shared Churches Project, which investigates religious coexistence and territorial dynamics in early modern Central Europe.

*Interior West Consulting | **GIS Analyst***

May 2020 - Apr 2024

Managed geospatial data for archaeological surveys across Colorado, New Mexico, and Wyoming. Processed GPS field data and integrated cultural resource datasets from NMCRIS and OAHP Compass into geodatabases using ArcMap and ArcGIS Pro. Used Mapbox Studio to create field maps for archaeological crews. Created and maintained GIS layers for the San Juan National Forest and generated over 400 high-quality plan and location maps in ArcGIS Pro for federal agency reports (BLM, USFS). Developed custom Python scripts using ArcPy to automate the creation of topographic vector layers from ASTER DEMs and used R for raster manipulation and terrain modeling.

*Various Employers | **Mapping and Archaeological Crew Chief***

2004 - 2017

Led mapping operations for archaeological survey and excavation projects across the American Southwest, the Great Plains, northwestern Belize, and parts of Europe (Spain and France). Oversaw spatial data collection, topographic mapping, and site documentation for a range of prehistoric and historic cultural sites. Conducted high-accuracy total station mapping using Topcon, Leica, Nikon/TDS, and Sokkia/TDS equipment. Managed spatial datasets in both ArcGIS and QGIS, integrating field data into geodatabases to support reporting and analysis. Produced topographic and feature maps for Ancestral Pueblo settlements (including Pueblo Bonito), Jornada Mogollon villages, Classic Maya residences, and Paleolithic cave sites. Supervised and trained field crews and graduate students in mapping workflows, data recording, and field documentation standards. Coordinated with project directors, CRM firms, and university research teams on both small-scale feature mapping and large-scale excavations.

*Maness and Associates | **Surveyor's Assistant/Instrument Man***

Feb 2003 - Apr 2003

Instrument man for a PLS (Ernie Maness) running a Sokkia total station on several land surveying projects around Cortez, CO, including land title surveys, right-of-way stakeouts, etc.

*CAJE Engineering | **Surveyor's Assistant/Instrument Man***

Jan 2002 - Oct 2002

Instrument man for CAJE Engineering running a Sokkia total station on several land surveying projects around Montezuma County, CO, including land title surveys, right-of-way stakeouts, etc. This work included establishing a traffic intersection at County Road G in Cortez, CO, slope staking, and performing bench loops (Rich Feit, Party Chief).

EDUCATION

University of Arizona

Master of Science • GIS • 2024 – 2025 • GPA: 4.00

University of New Mexico

Master of Arts • Archaeology • GPA: 4.03

University of Alabama

Bachelor of Arts • Archaeology and History

SKILLS

GIS & Spatial Analysis

ArcGIS Pro/ArcMap · ArcGIS Online · ArcGIS Enterprise · QGIS · Mapbox Studio · ModelBuilder · Geodatabase Design · Raster Analysis · Terrain Modeling · Viewshed Analysis · Topographic Mapping · Digitizing & Georeferencing · Cartographic Design

Programming & Automation

Python (ArcPy, Pandas) · R (raster processing) · SQL · Scripting GIS Workflows · Batch Processing · Spatial Data QA/QC · HTML · CSS · JavaScript

Field Data Collection & Equipment

Total Station Mapping (Topcon, Leica, Nikon/TDS, Sokkia/TDS) · Mapbox Studio · Avenza · Trimble GPS · GPS Field Data Integration · NMCRIS & OAHF Compass Queries · Feature Mapping & Plan/Profile Drawings

Domain Expertise

Cultural Resource Management (CRM) · Archaeological GIS · Environmental & Compliance Mapping · Historical Site Documentation · Public Land Management Support · Federal & State Reporting

Communication & Collaboration

Technical Writing · Graduate Teaching Support (GIS) · Mentoring & Training · Interdisciplinary Teamwork · Spatial Data Interpretation for Stakeholders

REFERENCES

Mr. Mark Lowe | Supervisor, Interior West Consulting | (970) 769-6455 | mark.angelowe@yahoo.com

Dr. Yoga Korgaonkar | Professor of GIS, University of Arizona | (813) 405-6171 **OR** (520) 626-8107 | yoganandk@arizona.edu

Dr. Chris Lukinbeal | GIS Director/Shared Churches Project, University of Arizona | (520) 621-6181 | clukinbe@arizona.edu

Dr. Craig Wissler | Professor of GIS, University of Arizona | (520) 621-7255 | cwissler@cals.arizona.edu